Run report: what sets the MutantXL closure degree (ℓ, N or m·ℓ)

- **Study:** `research/dreg_closure_driver_20261016/`, pre-registered in `PREREGISTRATION.md` (commit `b85dd3226`).
- **Run:** 2026-10-16, 20:49 to 23:06 UTC.
- **Class:** stage diagnostic. No `S`, no rho ratio, no speed claim.

What ran

- **Draws.** `dreg_ladder` with seed 20261017 (exact solution count; XL built to degree 3 only).
  - Each cell's draws were committed before it was measured.
  - The first generation pass of (24,6) was cut off by my wrapper's 1,700 s limit and rerun. The three draws both passes produced are identical apart from the timing field.
- **Measurement.** One process per draw: `prlimit --as=15e9 timeout 7200 dreg_closure --to 4 --max-cols 155000 --draw K`.
- **Cells.** m = 3 (22,6) and (24,6) at N = 40 and 42, and m = 4 (11,5) and (12,5) at N = 42 and 44. Four unsatisfiable draws per cell, 16 in all.
- **Host.** The 4-core, 16 GB container, not isolated (`AGENTS.md` §10).

Completion

- **Processes.** 20 of 20 (16 draws and 4 controls) exited 0.
- **Closure.** Every closure saturated.
- **Peak RSS.** 7.8 GB. The slowest draw took 730 s.

Measured values

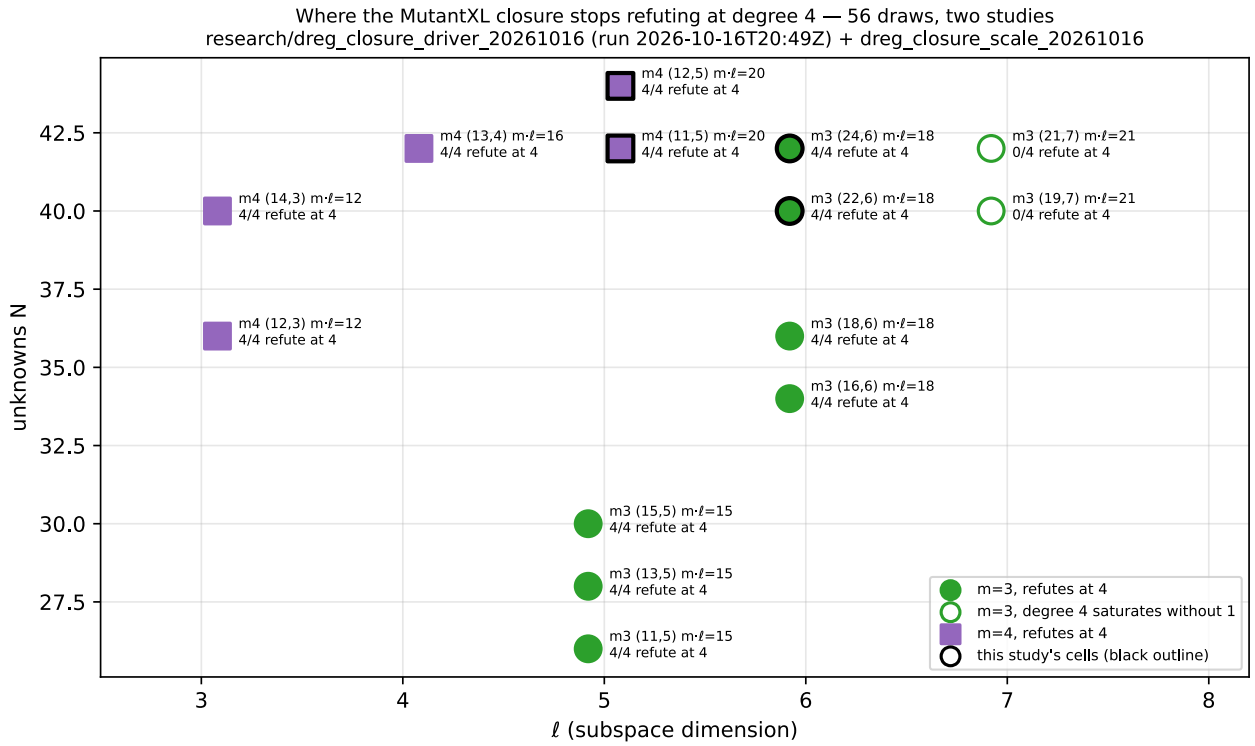


Figure 1: Recorded outcomes of this study (black outlines) and the scale study, by (ℓ, N). Filled markers refute at degree 4 on every draw. Hollow markers saturate at degree 4 without refuting on every draw. Source: `plot.py` (layout only).

- **Closure.** It refutes at degree 4 on 16 of 16 draws.
- **XL.** It refutes at degree 4 on none.
- **Work ratio.** $\log_2$ ops ÷ $\log_2$ cols is 2.18–2.33.
- **Controls.** Four satisfiable m = 4 draws, none refuted.
- **Per-draw values.** In `score-output.txt`.

Raw records

- `runs/*.d*.jsonl` and `.log`.
- `runs/queue.log`.

- `draws/*.jsonl`.
- Interpretation: `../../RESULTS.md`.